is now being planned. Planck, and CMBPol in particular, will not focus solely on temperature variations of the microwave background radiation, but will also measure *polarization*, the average spin directions of the microwave photons detected. Through a chain of reasoning too involved to cover here, it turns out that gravitational waves from the bang would leave a distinct imprint on the polarization of the microwave background radiation, perhaps an imprint large enough to be measured.

So, within a decade, we may get sharp insight into whether the bang was really a splat, and whether the universe we're aware of is really a three-brane. In the golden age of cosmology, some of the wildest ideas may actually be testable.

Dark Matter, Dark Energy, and the Future of the Universe

In Chapter 10 we went through the strong theoretical and observational evidence indicating that a mere 5 percent of the universe's heft comes from the constituents found in familiar matter—protons and neutrons (electrons account for less than .05 percent of ordinary matter's mass)—while 25 percent comes from dark matter and 70 percent from dark energy. But there is still significant uncertainty regarding the detailed identity of all this dark stuff. A natural guess is that the dark matter is also composed of protons and neutrons, ones that somehow avoided clumping together to form light-emitting stars. But another theoretical consideration makes this possibility very unlikely.

Through detailed observations, astronomers have a clear knowledge of the average relative abundances of light elements—hydrogen, helium, deuterium, and lithium—that are scattered throughout the cosmos. To a high degree of accuracy, the abundances agree with theoretical calculations of the processes believed to have synthesized these nuclei during the first few minutes of the universe. This agreement is one of the great successes of modern theoretical cosmology. However, these calculations assume that the bulk of the dark matter is *not* composed of protons and neutrons; if, on cosmological scales, protons and neutrons were a dominant constituent, the cosmic recipe is thrown off and the calculations yield results that are ruled out by observations.